SOF203 Fundamentals of Network Technology

Final Exam Revision

1. What are the five layers in the Internet protocol stack? What are the principal responsibilities of each of these layers?

2. What advantage does a circuit-switched network have over a packet-switched network?

3. Consider a packet of length 2,000 bytes to be sent over a link of distance 2,500 km, propagation speed 2.5×10^8 m/s, and transmission rate 100 Mbps.
   a. Calculate the propagation delay for this packet.
   b. Calculate the transmission delay for this packet.

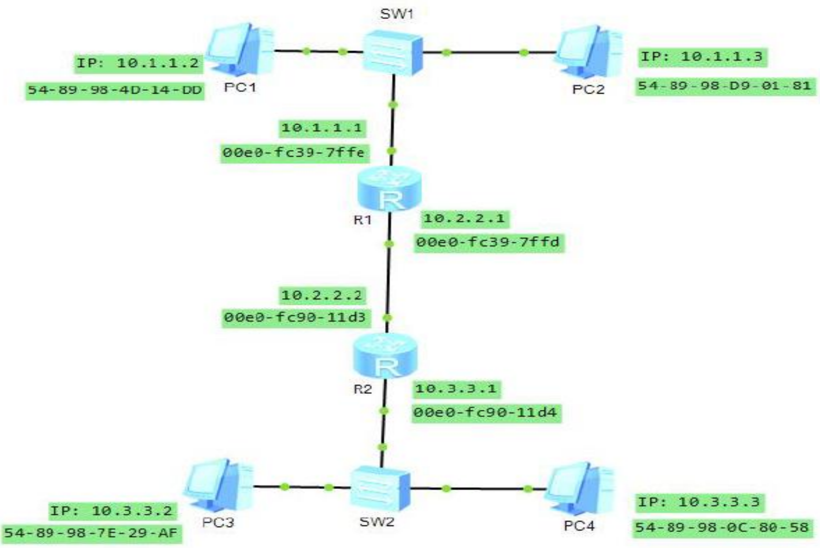
4. Why is an ARP query sent within a broadcast frame? Why is an ARP response sent within a frame with a specific destination MAC address?

5. MAC addresses and IP addresses for the interfaces at PC1, Router R1, Router R2, and PC3 are shown in Figure 3. Suppose PC1 sends a datagram to PC3.

   Give the source and destination MAC addresses in the frame encapsulating this IP datagram as the frame is transmitted

   i) from PC1 to R1,
   ii) from R1 to R2,
   iii) from R2 to PC3.

   Also give the source and destination IP addresses in the IP datagram encapsulated within the frame at each of these points in time.



6. Let's consider the operation of a learning switch in the context of a network in which 6 nodes labelled A through F are star connected into an Ethernet switch. Suppose that

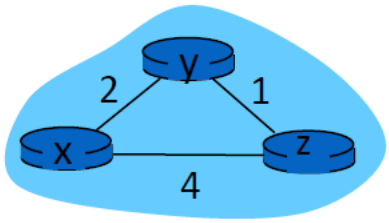
   i) B sends a frame to E,
   ii) E replies with a frame to B,
   iii) A sends a frame to B,
   iv) B replies with a frame to A.

   The switch table is initially empty. Show the state of the switch table before and after each of these events. For each of these events, identify the link(s) on which the transmitted frame will be forwarded, and briefly justify your answers.

7. An organization is granted the block 130.56.0.0/16. The administrator wants to create 1024 subnets. Show the subnet planning for the first four subnets.

Subnet	**Network Address**	**First usable IP**	**Last usable IP**	**Broadcast Address**	**Subnet Mask**
1					
2					
3					
4					

8. Consider the three-node topology shown in Figure Q2. The link costs are c(x,y) = 2, c(y,z) = 1, c(z,x) = 4. Compute the distance tables after the initialization step and after each iteration of a synchronous version of the distance-vector algorithm.



9. In a TCP connection, the initial sequence number at the client site is 3974188230. The client opens the connection with the server through 3-way handshake. The initial sequence number at the server site is 1884869393. Upon the completion of the 3-way handshake process, the client sends a data segment of 331 bytes. The server responds an acknowledgment upon receiving this data segment.

   i) Determine the acknowledgment number in the SYN/ACK segment sent by the server.
   ii) Determine the sequence number in the ACK segment sent by the client.
   iii) Determine the acknowledgment number in the ACK segment sent by the client.
   iv) What is the sequence number of the data segment sent by the client?
   v) What is the acknowledgment number in the acknowledgment of this data segment?

10. Consider the following network as shown in Figure Q1. With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the shortest path from t to all network nodes. Show how the algorithm works by filling in the following table.

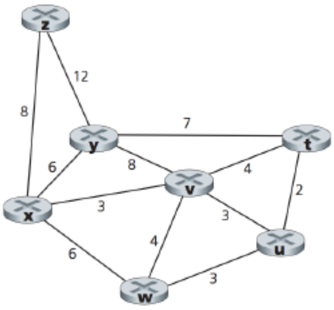


Figure Q1

N'	*D(u)*	*p(u)*	*D(v)*	*p(v)*	*D(w)*	*p(w)*	*D(x)*	*p(x)*	*D(y)*	*p(y)*	*D(z)*	*p(z)*

Notation:

D(v): cost of the least-cost path from the source node to destination *v* as of this iteration of the algorithm.
p(v): previous node (neighbor of *v*) along the current least-cost path from the source to *v*.
N': subset of nodes; *v* is in *N'* if the least-cost path from the source to *v* is definitively known.

11. Host A and B are communicating over a TCP connection, and Host B has already received from A all bytes up through byte 126. Suppose Host A then sends two segments to Host B back-to-back. The first and second segments contain 80 and 40 bytes of data, respectively. In the first segment, the sequence number is 127, the source port number is 302, and the destination port number is 80. Host B sends an acknowledgment whenever it receives a segment from Host A.

i) In the second segment sent from Host A to B, what are the sequence number, source port number, and destination port number?
ii) If the first segment arrives before the second segment, in the acknowledgment of the first arriving segment, what is the acknowledgment number, the source port number, and the destination port number?

iii) If the second segment arrives before the first segment, in the acknowledgment of the first arriving segment, what is the acknowledgment number?

12. Consider the following string of ASCII characters that were captured by Wireshark when the browser sent an HTTP GET message. Answer the following questions, indicating where in the HTTP GET message below you find the answer.

```
GET /cs453/index.html HTTP/1.1
Host: gaia.cs.umass.edu
User-Agent: Mozilla/5.0 (Windows;U; Windows NT 5.1; en-US; rv:1.7.2)
Gecko/20040804 Netscape/7.2 (ax)
Accept:ext/xml, application/xml, application/xhtml+xml,
text/html;q=0.9, text/plain;q=0.8,image/png,*/*;q=0.5
Accept-Language: en-us,en;q=0.5
Accept-Encoding: zip,deflate
Accept-Charset: ISO-8859-1,utf-8;q=0.7,*;q=0.7
Keep-Alive: 300
Connection:keep-alive
```

i) What is the URL of the document requested by the browser?
ii) What version of HTTP is the browser running?
iii) Does the browser request a non-persistent or a persistent connection?
iv) What type of browser initiates this message? Why is the browser type needed in an HTTP request message?

13. The text below shows the reply sent from the server in response to the HTTP GET message in the question above. Answer the following questions, indicating where in the message below you find the answer.

```
HTTP/1.1 200 OK
Date: Tue, 07 Mar 2008 12:39:45GMT
Server: Apache/2.0.52 (Fedora)
Last-Modified: Sat, 10 Dec2005 18:27:46 GMT
ETag: "526c3-f22-a88a4c80"
Accept- Ranges: bytes
Content-Length: 3874
Keep-Alive: timeout=max=100
Connection: Keep-Alive
Content-Type: text/html; charset= ISO-8859-1

<!doctype html public "- //w3c//dtd html 4.0
transitional//en">
<html>
<head>
<meta http-equiv="Content-Type" content="text/html;
charset=iso-8859-1">
<meta name="GENERATOR" content="Mozilla/4.79 [en] (Windows
NT 5.0; U) Netscape]">
<title>CMPSCI 453 / 591 / NTU-ST550A Spring 2005
homepage</title>
</head>
<much more document text following here (not shown)>
```

i) Was the server able to successfully find the document or not? What time was the document reply provided?
ii) When was the document last modified?
iii) How many bytes are there in the document being returned?
iv) What are the first 5 bytes of the document being returned?